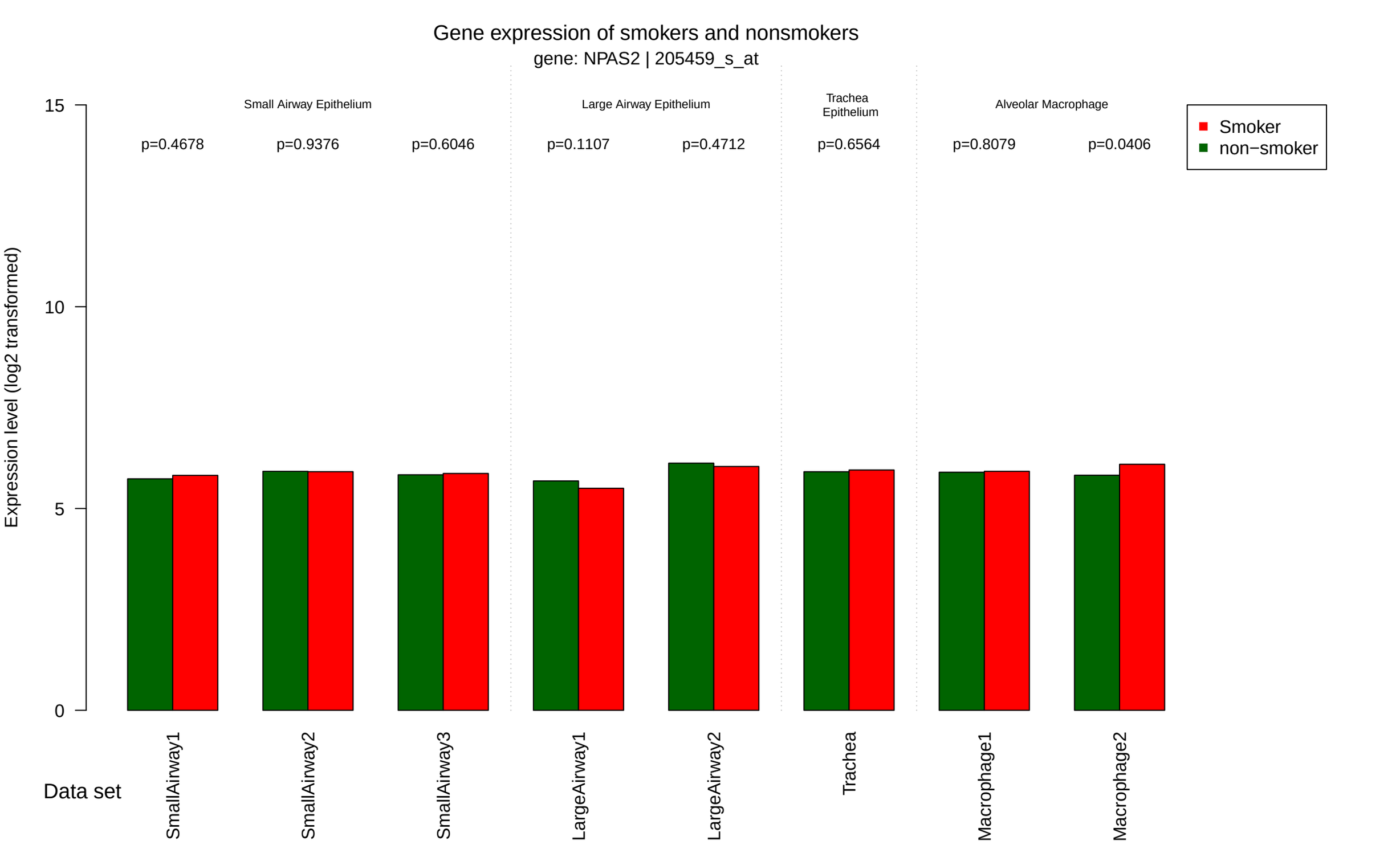
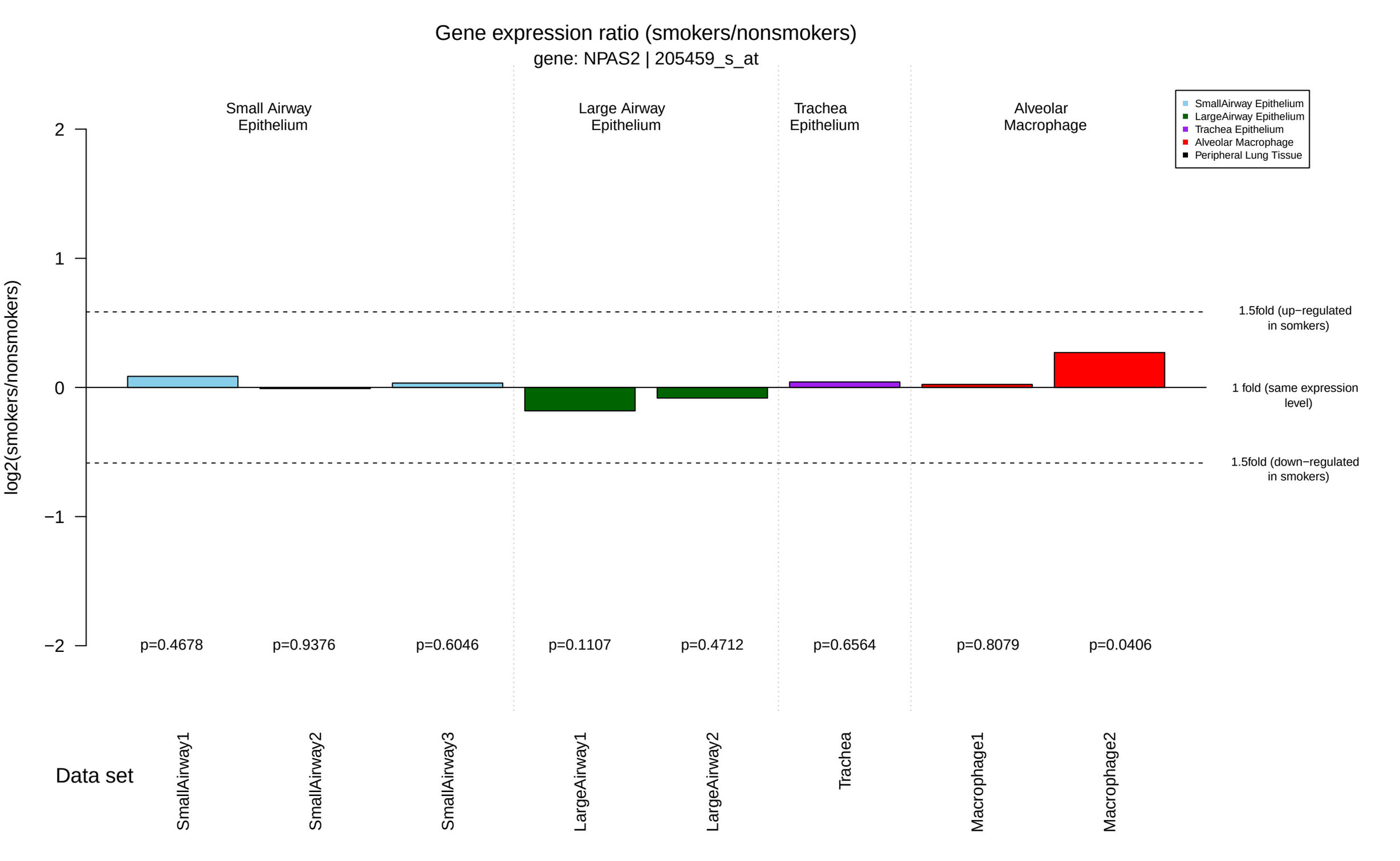
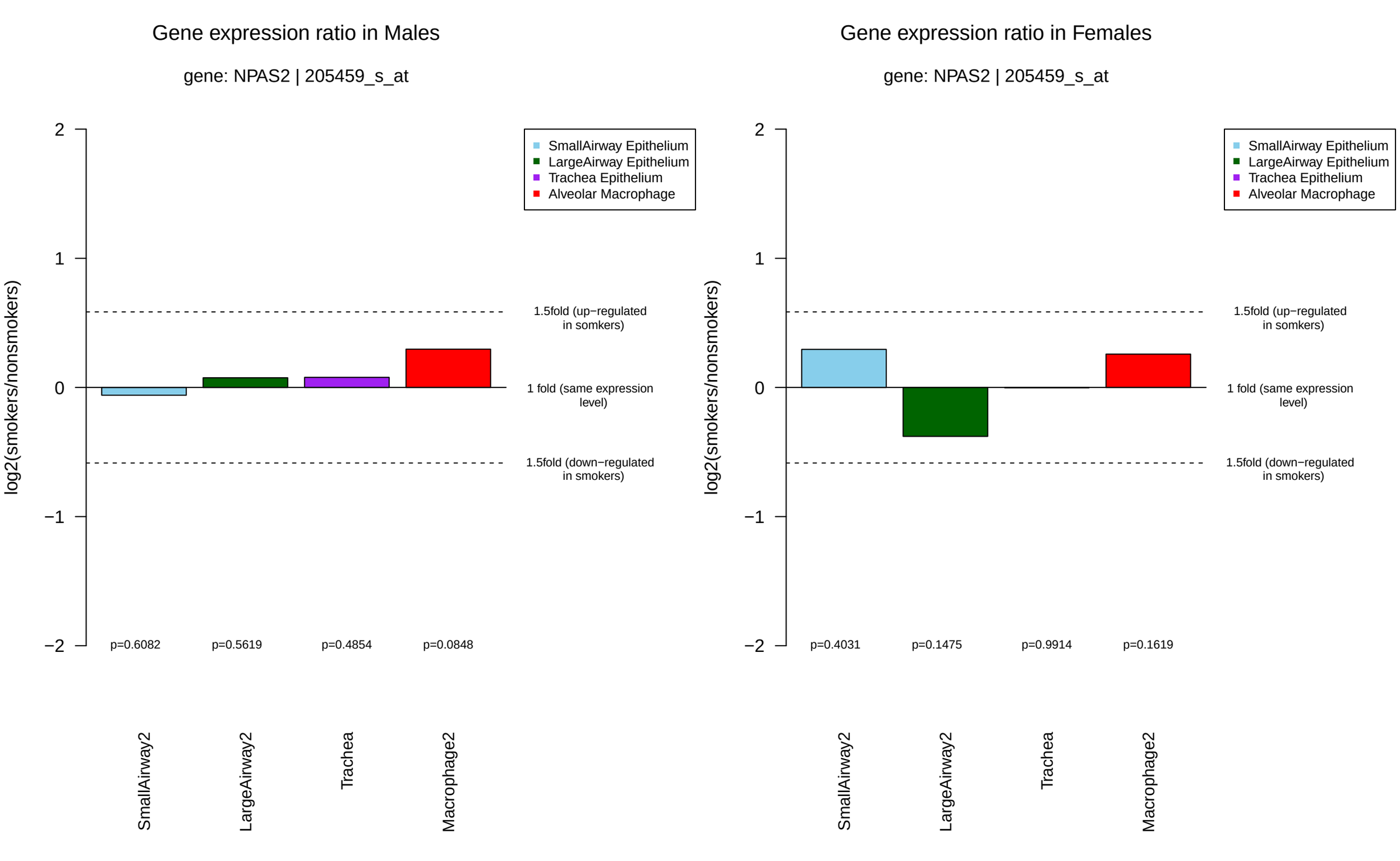
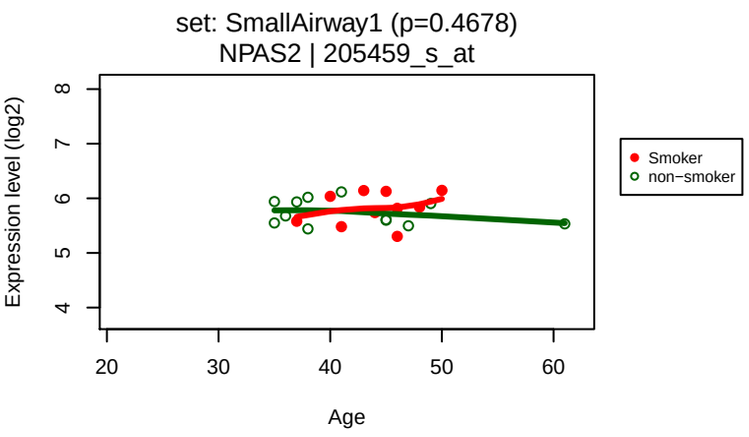
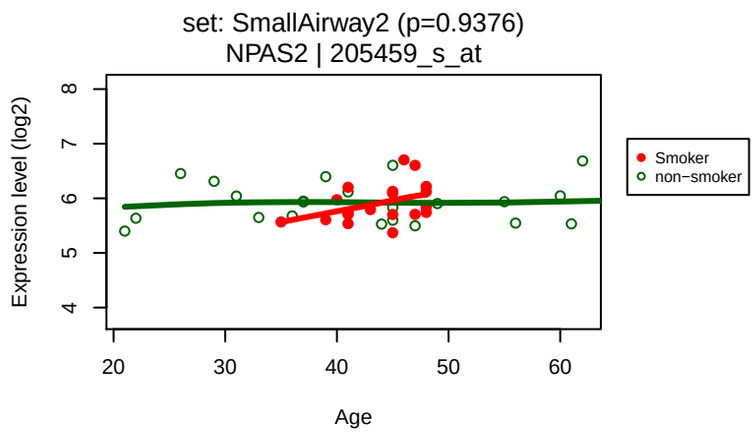
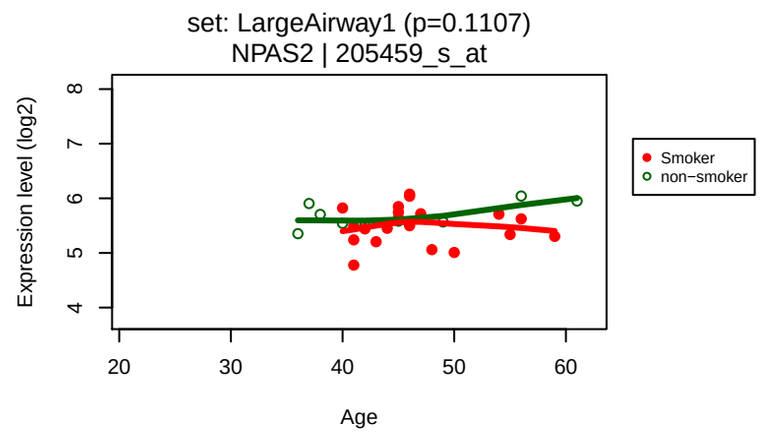

Gene expression of smokers and nonsmokers
gene: NPAS2 | 205459_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4678
p=0.9376
p=0.6046
p=0.1107
p=0.4712
p=0.6564
p=0.8079
p=0.0406
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

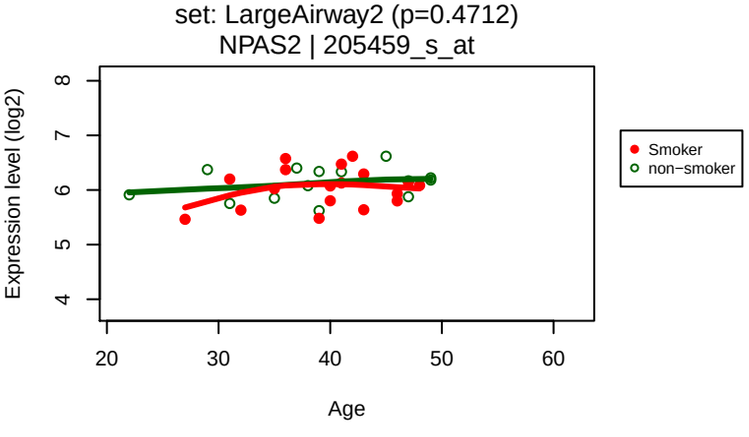

Gene expression ratio (smokers/nonsmokers)
gene: NPAS2 | 205459_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4678
p=0.9376
p=0.6046
p=0.1107
p=0.4712
p=0.6564
p=0.8079
p=0.0406
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

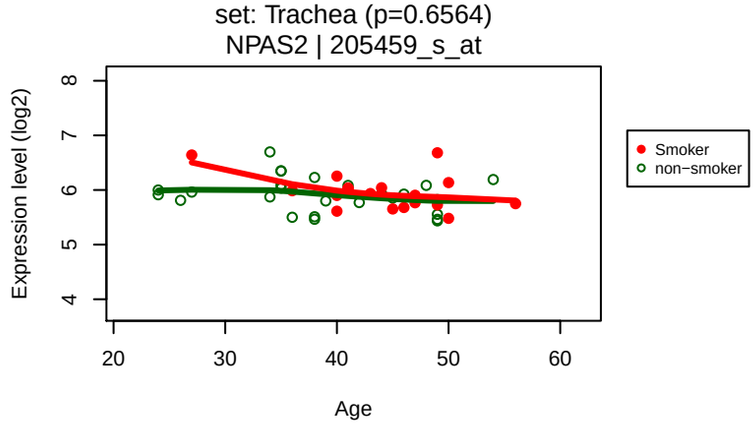

Gene expression ratio in Males
gene: NPAS2 | 205459_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6082
p=0.5619
p=0.4854
p=0.0848
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS2 | 205459_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4031
p=0.1475
p=0.9914
p=0.1619
SmallAirway2
LargeAirway2
Trachea
Macrophage2

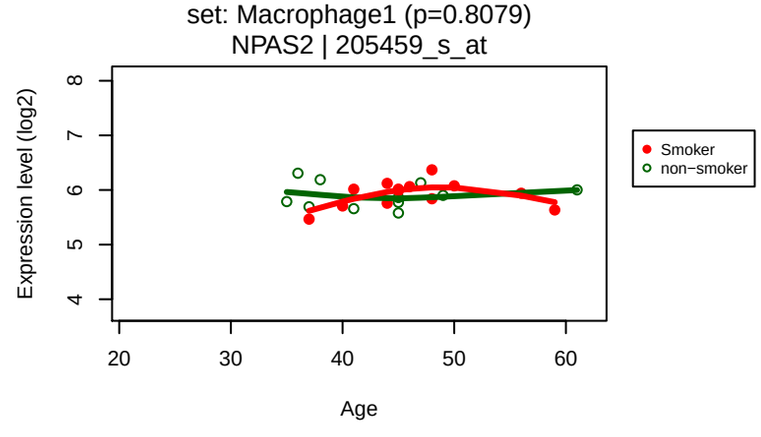

set: SmallAirway1 (p=0.4678)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker

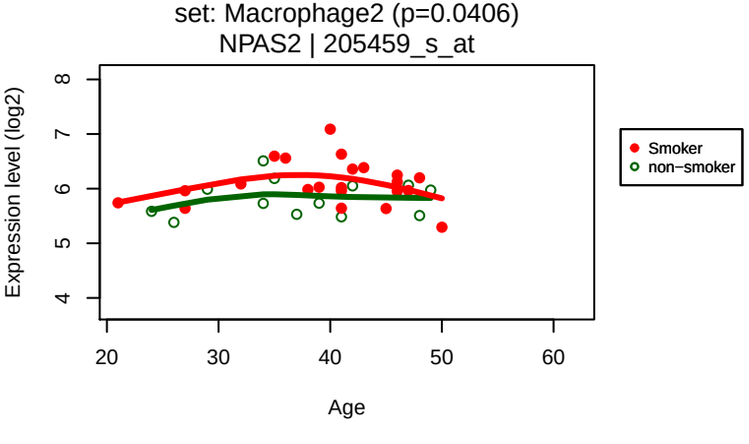

set: SmallAirway2 (p=0.9376)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker

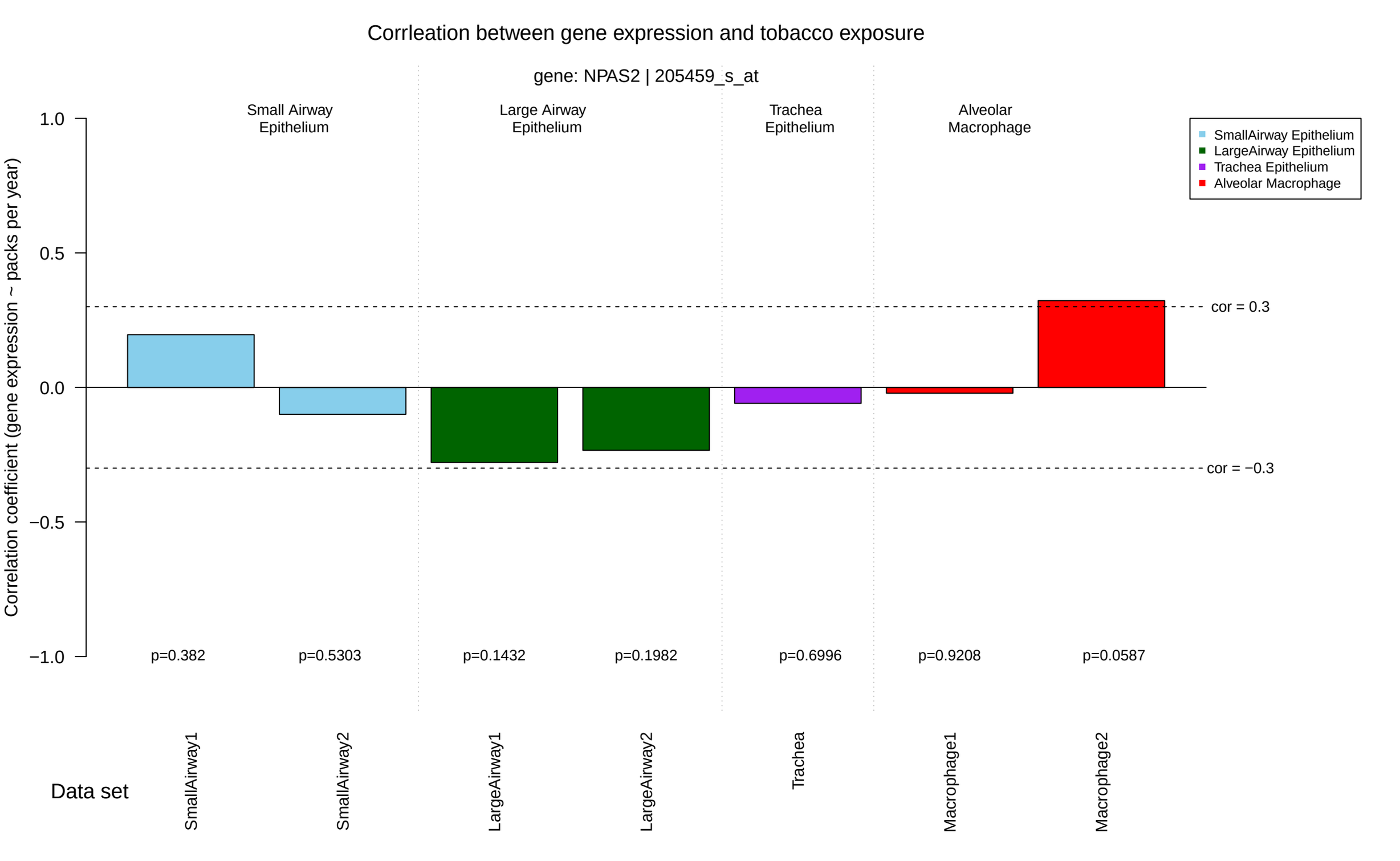

set: LargeAirway1 (p=0.1107)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4712)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6564)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8079)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0406)
NPAS2 | 205459_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS2 | 205459_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.382
p=0.5303
p=0.1432
p=0.1982
p=0.6996
p=0.9208
p=0.0587
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2